Ajay Mandlekar
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Overview

I am a research scientist at NVIDIA AI working with Dieter Fox in the NVIDIA Seattle Robotics Lab. I obtained my PhD degree from Stanford University, co-advised by Silvio Savarese and Fei-Fei Li. My research focuses on developing systems and algorithms to enable robots to learn manipulation tasks.

Selected Publications (see website for full list)

- *Learning and Retrieval from Prior Data for Skill-based Imitation Learning*
  Soroush Nasiriany, Tian Gao, **Ajay Mandlekar**, Yuke Zhu
  CoRL 2022
  [paper] [website]

- *MoCoDA: Model-based Counterfactual Data Augmentation*
  Silviu Pitis, Elliot Creager, **Ajay Mandlekar**, Animesh Garg
  NeurIPS 2022
  [paper] [website] [code]

- *MineDojo: Building Open-Ended Embodied Agents with Internet-Scale Knowledge*
  Linxi Fan, Guanzhi Wang*, Yunfan Jiang*, **Ajay Mandlekar**, Yuncong Yang, Haoyi Zhu, Andrew Tang, De-An Huang, Yuke Zhu, Anima Anandkumar
  NeurIPS 2022 Datasets and Benchmarks Track
  [paper] [website] [code] [blog post]

- *What Matters in Learning from Offline Human Demonstrations for Robot Manipulation*
  Ajay Mandlekar, Danfei Xu, Josiah Wong, Soroush Nasiriany, Chen Wang, Rohun Kulkarni, Li Fei-Fei, Silvio Savarese, Yuke Zhu, Roberto Martin-Martin
  CoRL 2021
  Oral (6.5% Acceptance)
  [paper] [website] [video] [blog post] [code]

- *Error-Aware Imitation Learning from Teleoperation Data for Mobile Manipulation*
  Josiah Wong, Albert Tung, Andrey Kurenkov, **Ajay Mandlekar**, Li Fei-Fei, Silvio Savarese, Roberto Martin-Martin
  CoRL 2021
  [paper] [website] [video]

- *Human-in-the-Loop Imitation Learning using Remote Teleoperation*
  Ajay Mandlekar, Danfei Xu*, Roberto Martin-Martin*, Yuke Zhu, Li Fei-Fei, Silvio Savarese
  Preprint
  [paper] [website] [video]

- *Generalization through Hand-Eye Coordination: An Action Space for Learning Spatially-Invariant Visuomotor Control*
  Chen Wang*, Rui Wang*, **Ajay Mandlekar**, Li Fei-Fei, Silvio Savarese, Danfei Xu
  IROS 2021
  [paper] [website]

- *Learning Multi-Arm Manipulation through Collaborative Teleoperation*
  Albert Tung*, Josiah Wong*, **Ajay Mandlekar**, Roberto Martin-Martin, Yuke Zhu, Li Fei-Fei, Silvio Savarese
  ICRA 2021
  Best Multi-Robotic Systems Paper Finalist
  [paper] [website] [video]

- *Deep Affordance Foresight: Planning Through What Can Be Done in the Future*
  Danfei Xu, **Ajay Mandlekar**, Roberto Martin-Martin, Yuke Zhu, Silvio Savarese, Li Fei-Fei
  ICRA 2021
  [paper] [website] [video]

- *robosuite: A Modular Simulation Framework and Benchmark for Robot Learning*
  Yuke Zhu, Josiah Wong, **Ajay Mandlekar**, Roberto Martin-Martin
  Technical Report
  [paper] [website] [code]

- *Learning to Generalize Across Long-Horizon Tasks from Human Demonstrations*
  Ajay Mandlekar*, Danfei Xu*, Roberto Martin-Martin, Silvio Savarese, Li Fei-Fei
  RSS 2020
  [paper] [website] [blog post] [video]

- *IRIS: Implicit Reinforcement without Interaction at Scale for Learning Control from Offline Robot Manipulation Data*
  Ajay Mandlekar, Fabio Ramos, Byron Boots, Silvio Savarese, Li Fei-Fei, Animesh Garg, Dieter Fox
  ICRA 2020
  [paper] [website] [video]

- *Scaling Robot Supervision to Hundreds of Hours with RoboTurk: Robotic Manipulation Dataset through Human Reasoning and Dexterity*
  Ajay Mandlekar, Jonathan Booher, Max Spero, Albert Tung, Anchit Gupta, Yuke Zhu, Animesh Garg, Silvio Savarese, Li Fei-Fei
  IROS 2019
  Best Cognitive Robotics Paper Finalist
  [paper] [website] [blog post]

- *RoboTurk: A Crowdsourcing Platform for Robotic Skill Learning through Imitation*
  Ajay Mandlekar, Yuke Zhu, Animesh Garg, Jonathan Booher, Max Spero, Albert Tung, Julian Gao, John Emmons, Anchit Gupta, Emre Orbay, Silvio Savarese, Li Fei-Fei
  CoRL 2018
  [paper] [website] [video]

Community Service

Workshop Organizer
- Co-organized Overlooked Aspects of Imitation Learning: Systems, Data, Tasks, and Beyond workshop at RSS 2022.
- Co-organized Advances and Challenges in Imitation Learning for Robotics workshop at RSS 2020.

Peer Review
- Reviewer for NeurIPS, ICML, RSS, CoRL, IROS, CVPR, IEEE T-RO.

Education

Stanford University
NDSEG Fellow
PhD student, Electrical Engineering
M.S. Computer Science, Class of 2018
GPA: 4.1

California Institute of Technology
B.S. Electrical Engineering, B.S. Computer Science
Class of 2016
Officer of Tau Beta Pi Engineering Honor Society
GPA: 4.1
Programming Languages: Python, C, Matlab, R, OCaml, Haskell, Swift, IA32, x86, VHDL

Work Experience

NVIDIA Seattle Robotics Lab Summer 2019
Research Intern
- Developed IRIS algorithm for learning from offline crowdsourced robot manipulation data. Supervised by Dieter Fox.

Stanford Vision and Learning Lab Winter 2017 - Present
Stanford EE PhD
- Advised by Silvio Savarese and Fei-Fei Li

Apple Summer 2015
Software Engineering Intern on iOS Location and Motion
- Implemented some deep learning methods and presented to Craig Federighi, the SVP of Software.

Caltech High-Speed Integrated Circuits Summer 2014
Research Intern, Robert J. McEliece and David Rutledge SURF Fellow
- Spent a summer in Professor Ali Hajimiri's lab. Implemented a wireless feedback mechanism using a Raspberry Pi, Android tablet, Arduino Nano.
- Designed both hardware (PCB Design, digital and analog circuit development) and software (control, sensing, and optimization) interfaces and implemented wireless and serial communication protocols.

SLAC National Accelerator Laboratory Summer 2013
Research Intern
- Studied accelerator physics relating to the operation of the Free Electron Laser and the production of extremely high-power x-rays used by biologists.
- Investigated the theory of harmonic lasing in order to try to make the shift towards shorter wavelength, higher energy radiation.

Teaching

- CS 332 - Advanced Survey of Reinforcement Learning (Teaching Assistant - Stanford)
- CS 231N - Convolutional Neural Networks for Visual Recognition (Teaching Assistant - Stanford)
- ACM 95ab - Complex Analysis, Differential Equations (Teaching Assistant - Caltech)
- CS 24 - Introduction to Computing Systems (Teaching Assistant - Caltech)

Selected Coursework

- CS 124 - Operating Systems
  - Built a command shell, bootloader, priority scheduler, MLFQ scheduler on top of Pintos.
- CS 155/156 - Machine Learning and Data Mining
  - Netflix Challenge - placed 3rd in class using a blend of Time-SVD++, RBM, and kNN models.
- EE 52/90/91 - Embedded Systems Hardware, Analog Electronics Projects
  - PCB Design, built an MP3 player, a “smart” dog bowl and a function generator.
- EE 111/112 - Digital Signal Processing
  - DSP Systems, Difference Equations, Z-Transform, Multirate Systems, Filter Banks, Filter Design.
- EE 113/114 - Feedback and Control Circuits, Analog Circuit Design
  - Basic feedback circuits, compensation, PID, digital control, amplifier design, SPICE simulation.
- EE 189 - Design and Construction of Biodevices
  - Built a pulse oximeter and a PCR machine.
- EE 364ab - Convex Optimization